C64uRemote for M5Dial

User Manual

Version 1.0

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Welcome

C64uRemote turns an **M5Dial** into a convenient remote control for the **Commodore 64 Ultimate (c64u)** and the **Ultimate64 Elite-II**. Over Wi-Fi you can reset and reboot the machine, power it off, open the Ultimate menu and change the CPU speed.

The M5Dial ships with an **NFC reader built in**. Together with a retrofitted **microSD card** it becomes a tap-to-play console: hold an NFC tag against the device and the matching game starts on the C64. The tags use the TeensyROM and Zaparoo format, so the same tag works on those systems and on the M5Stack Core version of this project.

This project builds on the original idea by Karl Prosser (@klumsy) and was extended for the M5Dial by Martin Oswald (@mad, 1MHz.de).

What you need

Required:

- An M5Dial
- A Commodore 64 Ultimate or Ultimate64 Elite-II on the same Wi-Fi network

Optional, for launching games from storage:

- A microSD module wired to Port A and Port B (installation is covered in the technical documentation)
- A microSD card formatted as FAT32
- For cordless use: a 3.7 V Li-ion/LiPo battery with a 1.25 mm plug
- NFC tags: NTAG215 recommended, NTAG213/216 and MIFARE Classic also work

Without an SD card every other function still works. The NFC reader is always present, but without an SD card it has nothing to launch.

First-time setup

Installing the firmware – without a development environment

The ready-made firmware can be installed straight from the browser, with no VS Code or PlatformIO: open **https://1mhz.de/M5-webflash/** in **Chrome** or **Edge** on a computer (Safari and Firefox cannot do USB), connect the M5Dial with a USB **data** cable, pick device and language and click *Install*. The same page also installs every later update.

The firmware from the page contains no credentials. Set up WiFi and the c64u address on the device afterwards as described below. If the connection fails, the hints on the page will help.

For the M5Dial to find your C64, the Wi-Fi credentials and the address of the C64 have to be stored once. You can do that **right on the device** under *Settings → WiFi* – the source code no

longer has to be touched. The credentials go into internal memory and survive every restart. The *Setting up Wi-Fi* chapter walks through the details.

If you would rather supply the credentials while programming the device, put them into `build_env.h` as before (see the technical documentation). They then serve as the initial values for the very first start.

Connection status is always visible as a small **dot at the very top** of the round display. In the ring menu the *c64u Power Off* icon sits there, so the **Status icon (i)** carries the colour instead – the glyph itself and, while it is not selected, its frame as well.

Dot	Meaning
green	Everything connected, ready
blue	Wi-Fi up, but the C64 does not answer
yellow	C64 reachable, but the password is wrong
red	No Wi-Fi connection

At the bottom edge of every menu page you see two labels: **NFC** and **SD**. Green means detected, grey means not present. Full details are on the *Status* page.

Operation

The M5Dial has three inputs: the **rotary bezel**, the **button** (press the bezel down) and the **touchscreen**.

Input	Effect
Rotate	On the home screen: open the menu. Otherwise: move the selection or scroll
Short press	Select, confirm, execute
Long press (from 0.6 s)	Go back one level. On the home screen: shortcut action
Touch the ring	Run that menu entry directly
Touch the centre	Return from the menu to the home screen
Touch a list row	Select the row, a second tap runs it
Touch the title line	Go back
Long touch (home screen)	Second shortcut action

You do not have to hit the icons precisely: the whole outer ring is active and the device picks the icon closest to your finger. Only the centre with the logo returns to the home screen.

The home screen

The home screen shows the 1MHz logo, either still or with changing effects (*Water*, *RotoZoom*, *SineWave*, *Ripple*, *Raster*). It doubles as the screen saver: after the configured idle time the device returns here on its own.

A turn, a button press or a tap opens the menu.

Two shortcuts that skip the menu: on the home screen a long button press runs a freely selectable action (*Reset* by default) and a long touch on the display runs a second one (*Ultimate Menu* by default). Both are configured under *Settings*. The long touch also works inside the ring menu, so you do not have to return to the home screen first.

The menu

Ten icons sit in a circle with the logo in the middle. The selected icon is larger and brightly outlined; its name appears briefly above the logo.

Icon	What it does
c64u Power Off	Powers the C64 down – press twice for safety. **Holding it (1.5 s)** switches the M5Dial off instead
Reset	Resets the C64 (like the reset button)
Reboot	Restarts the C64 completely
Ultimate Menu	Opens or closes the Ultimate menu on the C64
CPU Speed	View and change the CPU speed
RFID / NFC	Present a tag and launch the stored game
SD card	Pick a game straight from the SD card and launch it
Joystick Swap	Swaps the joystick ports on the C64 (Normal ↔ Swapped)
Status	Detailed connection information, firmware version at the bottom
Settings	Preferences and NFC tools

Powering off the C64 (c64u Power Off): switching the machine off by accident would be annoying, so the prompt *c64u OFF? AGAIN!* appears first. Only a second press within the time window actually powers down.

Switching the M5Dial off: on the same icon, **hold** the button – or your finger on the icon – **for 1.5 seconds**. *OFF* appears; once you let go the M5Dial is off. Let go earlier and it stays the normal c64u Power Off with prompt.

Changing the CPU speed

Choose **CPU Speed** from the menu. The current speed is shown at the top, the list of options below. Turn the bezel to pick a speed, press to apply. The M5Dial reads the available steps straight from the C64, so you get exactly the values your machine supports.

Swapping the joystick ports

Some games expect the joystick in port 1, others in port 2. Instead of moving the cable, the mapping can be swapped inside the C64.

Choose **Joystick Swap** from the menu – every press toggles between *Normal* and *Swapped*, and *JOY Swapped* or *JOY Normal* appears briefly.

Settings → Joystick shows the current state and steps through every value your C64 offers: besides *Normal* and *Swapped* there may be *WASD P1* and *WASD P2*, depending on the firmware – the keyboard then drives that port.

The Ultimate firmware has no dedicated remote command for this. The M5Dial sets the *Joystick Swapper* item in the C64 configuration, exactly like the CPU speed. The state therefore survives until it is changed again, a reset included.

Launching games from a tag

Prerequisites: a microSD card with your games is installed, and the tag has been written beforehand (see below).

Just present the tag (automatic)

Normally you do not have to operate anything at all: while the home screen or the menu is showing, the M5Dial checks in the background whether a tag is present. As soon as it finds one it switches to reading mode on its own and launches the stored program.

The reading screen then stays up, so you can present the next tag right away. If nothing happens for a while, the device returns to the home screen.

How often it looks is set under *Settings → Auto-NFC*:

Setting	Meaning
Off	no background polling, tags only via the menu entry
1.5s	very frugal
0.7s	factory default, a good compromise
0.3s	quickest to react

The probe is short enough that even at *0.3s* it slows neither the animation nor the controls noticeably.

Through the menu entry

If you want to read a tag deliberately — or if *Auto-NFC* is switched off — the manual route still works:

1. Choose **RFID / NFC** from the menu.
2. Hold the NFC tag against the front of the M5Dial.
3. The device reads the stored path, fetches the file from the SD card and sends it to the C64. A progress bar shows the upload.
4. Once *START: …* appears (or *LAEUFT: …* for disk images), the game is running.

Take the tag away, present the next one – the screen stays in reading mode, so you can play several tags in a row.

Supported file types

Extension	What it is
`.prg`	Program (loaded and started)
`.crt`	Cartridge
`.sid`	Music file (played back)
`.mod`	Amiga music module
`.d64 .d71 .d81`	Disk image
`.g64 .g71`	Disk image (GCR format)

Disk images are mounted into the drive. What happens next is set under *Settings → Disk Action*: mount only, mount and reset, or mount, reset and load and run the first program.

Random launch

If the tag holds a `?` instead of a file name, or just a folder path, the M5Dial picks a random file from that folder every time. Handy for a “surprise tag”.

Command cards

A card does not have to point at a game – it can also carry a **command**. Present it and the command runs immediately, with no menu and no SD card needed.

Card	Effect
Reset	Resets the C64
Reboot	Restarts the C64 completely
Ultimate Menu	Opens or closes the Ultimate menu
c64u Off direct	Powers the C64 off immediately

Card	Effect
c64u Off with prompt	Asks first – present the same card a second time within the time window to confirm
M5Dial Power Off	Switches the M5Dial itself off
CPU x MHz	Sets the CPU to the value stored on the card
Swap Joystick	Swaps the joystick ports (Normal ↔ Swapped)
Joystick Normal / Swapped / WASD P1 / WASD P2	Sets the port mapping to that fixed value

Creating a command card

1. Choose **NFC-Cmd** in the settings.
2. Pick the command from the list. After the fixed entries come the joystick mappings and then all CPU steps your C64 offers, so a “CPU 10 MHz” card is a single click. *JOY* or *CPU* on the right tells the two blocks apart.
3. Present the card; *KARTE OK* means written and verified.

PowerOff with prompt

The waiting time lives **on the card**, not in the device. When writing, the value comes from *NFC-Cmd PowOff* (3, 5, 8 or 15 seconds, 8 s by default). Presenting such a card shows *c64u OFF? AGAIN!* with a countdown. To power off:

- present the **same card** again, or
- press the **button**

If the countdown expires or a different card appears, nothing happens. A card with a time of **0** powers off immediately without asking.

Switching the M5Dial off by card

A card holding `CMD:M5OFF` switches the **M5Dial itself** off – without a prompt, since presenting a card is already a deliberate act. So that a card still lying on the device at power-up does not switch it straight off again, it is ignored for the first eight seconds after start (*REMOVE CARD*). The same card also switches a Core, CoreS3 or Stick off.

What is stored on the card

The command is plain text in an NDEF record – you can inspect or write it with any NFC app on your phone:

```
CMD:RESET
CMD:REBOOT
CMD:MENU
CMD:POWEROFF=0      power off immediately
CMD:POWEROFF=8      ask first, 8 seconds to confirm
CMD:M5OFF           switch the M5Dial itself off
CMD:CPU=10          set the CPU to 10 MHz
CMD:JOY             toggle the joystick ports
CMD:JOY=SWAPPED     set the ports fixed; also NORMAL, WASD1, WASD2
CMD:DIRECT=192.168.4  switch direct mode on with this network
CMD:DIRECT=OFF      direct mode off, back to the stored WiFi
```

Case does not matter. The M5Dial and M5Stack Core editions understand the same format, so one card works on both.

Launching games straight from the SD card

It works the same way without a tag: choose **SD card** from the menu. You see the contents of the card, filtered down to launchable files.

- **Rotate** scrolls the list
- **Short press** opens a folder or launches a file
- **Long press** goes up one folder, and back to the menu at the top level
- `..` in the list also goes up one folder

To the right of each entry you see the file extension, or `dir` for folders.

Writing and managing NFC tags

The four NFC tools sit at the top of **Settings**.

NFC-Write: put a game on a tag

1. Choose **Settings → NFC-Write**.
2. Pick the game in the file browser.
3. Hold the NFC tag against the M5Dial.
4. *KARTE OK* means: written and verified.

The path is stored as an NDEF text record, the same format TeensyROM and Zaparoo use.

NFC-Info: read a tag

Choose **Settings → NFC-Info** and present a tag. You get the UID, tag type, memory size, the stored text, the path derived from it and whether that file actually exists on the SD card. Below that follow the raw contents of the tag.

Turn the bezel to scroll the list; tapping the upper or lower half also jumps.

NFC-Dump and NFC-Restore: copying tags

- **NFC-Dump** reads the tag and stores it as a text file at `/NFC-DUMPS/<UID>.nfc` on the SD card.
- **NFC-Restore** writes such a file back to a different tag. Pick the dump file first, then present the target tag.

Important: the UID of a tag is burned into the chip and cannot be copied. The *content* is copied – which is what matters here. Areas that could lock a tag permanently are deliberately left untouched.

Setting up Wi-Fi

Everything for this sits under **Settings → WiFi**. The M5Dial remembers up to **four networks** and tries them one after another when connecting – handy if you move between home and a phone hotspot.

Entry	Effect
Direct mode	Own WiFi without a router, the c64u connects directly (see chapter *Direct mode*)
Direct net	Address range of the direct network: `192.168.4.x` or `192.168.2.x`
Scan networks	Search the area and pick a network from the list
Load from SD	Read `wifi.txt` from the microSD
Setup portal	Access point of its own with a web interface
Saved	Pick a stored network and connect

Entry	Effect
To NFC card	Write a stored network to an NFC card
Save to SD	Write all stored networks to the microSD as `wifi.txt`
Delete network	Remove a single entry
Delete all	Discard all credentials

Route 1: scan, then hand over the password on a tag

Scan networks shows the networks found after a few seconds, along with their signal strength. A network that is already stored is marked *known* and connects straight away; an *open* network needs no password.

For everything else the M5Dial asks for the password and waits for an NFC tag. The tag carries an ordinary text record in the same scheme that Wi-Fi QR codes use:

```
WIFI:S:MyWiFi;T:WPA;P:MyPassword;;
```

Write the tag with a phone, for example with the **NFC Tools** app (*Write → Add a record → Text*). If the tag holds only the password without the `WIFI:` prefix, it is assigned to the network you picked beforehand.

> The password sits unencrypted on the tag. Once you are set up, overwrite the tag – the device keeps the password in internal memory anyway.

Presenting a tag is enough

A complete `WIFI:` tag also works **outside** the setup: present it on the home screen or in the ring menu and the M5Dial stores the network and connects to it right away. It waits up to eight seconds and then reports *WIFI ACTIVE* with the IP address, or *NETWORK NOT THERE* if the network is out of range. If that network is already connected, nothing happens (*ALREADY CONNECTED*) – so the tag may simply stay on the reader.

That gives a second device its credentials in a matter of seconds.

Route 2: a file on the SD card

Put a text file `wifi.txt` in the root directory of the microSD:

```
ssid = MyWiFi
pass = MyWiFiPassword

ssid = Hotspot
pass = secret123

host     = 192.168.0.64
hostpass =
```

Every new `ssid` line begins a new entry; lines starting with `#` are comments. The file is read at start-up (as long as no network is stored yet) and at any time through *Load from SD*.

The other way round, **Save to SD** writes all stored networks together with `host` and `hostpass` back out in exactly this format. An existing `wifi.txt` is renamed to `wifi.bak` first, so nothing is lost. That makes setting up a second device a no-typing affair – bear in mind that the passwords sit on the card in plain text.

Route 3: the setup portal

Setup portal turns the M5Dial into an access point of its own for five minutes. The display shows the network name, the password and the address you open in a browser. There you enter SSID and password – either from the list of networks found or by hand – and optionally the address and password of the C64.

After saving, the M5Dial shuts the access point down and connects to the new network. The browser connection breaking off in the process is normal.

Direct mode: no router, e.g. at meetings

At a meeting there is often no WiFi – or one you would rather not put the c64u on. For this case the M5Dial opens a small WiFi network of its own and the c64u connects to it directly. No router is needed.

Network name (SSID)	`C64uRemote-Direct`
Password	`c64ultimate`
M5Dial	`192.168.4.1`
c64u	`192.168.4.64`

Once on the c64u

The c64u can only remember **one** WiFi network. In the Ultimate menu, enter the direct network's name and password in the network/WiFi settings and leave address assignment on **DHCP**. Back home, enter your home network on the c64u again.

On the M5Dial

Under **Settings → WiFi**, select *Direct mode*. The M5Dial leaves its normal WiFi, starts the direct network and shows a page with everything the c64u needs: network name, password, the c64u's address and the connection state (*waiting for the c64u*, *c64u connected*).

As soon as the c64u joins, it gets the address `192.168.4.64` and the M5Dial checks right away whether it answers. After that everything works as usual. The c64u's stored home address is left untouched.

If another device (a phone, say) joins first, it gets `.64` and the c64u the next address. That is not a problem: the M5Dial then tries the connected devices one after another until the c64u answers. The address it found is shown on the direct mode page and in the status display.

Direct mode stays set after switching off – the M5Dial starts straight into the direct network next time.

Switching off: On the direct mode page, **press the button**. A tap on the screen only goes back to the WiFi menu; direct mode keeps running. The M5Dial then reconnects to the stored WiFi. The same happens when you pick a network under *Saved* or present a WiFi card.

Address range

Direct net switches between `192.168.4.x` and `192.168.2.x`. The M5Dial always has `.1`, the c64u `.64`. If direct mode is running, the network restarts right away; the c64u reconnects by itself.

For other values, `wifi.txt` on the SD card has lines of its own:

```
direct      = on                  on / off
direct_ssid = C64uRemote-Direct
direct_pass = c64ultimate         at least eight characters
direct_net  = 192.168.4           M5Dial = .1, c64u = .64
```

Save to SD writes these lines out as well.

Several devices at one table

If two remotes open a direct network at the same time, they need different network names – otherwise a c64u may join the wrong device. That is what *Direct no.* in the WiFi menu is for: every press steps from - (no number, `C64uRemote-Direct`) through `C64uRemote-Direct-1` to `-9` and back again. Each number also uses its own radio channel (1, 6 or 11) so the networks do not disturb each other.

On the c64u you enter exactly this network name – the direct mode page shows it. If direct mode is running, the network restarts with the new name right away. A name with a number in `wifi.txt` (`direct_ssid = C64uRemote-Direct-2`) works the same way; a completely custom name shows as *custom* and runs on channel 6.

Switching by card

NFC-Cmd offers three command cards for direct mode: first the network currently set, then the other one (`192.168.4.x` or `192.168.2.x`), last *Direct mode off*. The card then carries `CMD:DIRECT=192.168.4`, `CMD:DIRECT=192.168.2` or `CMD:DIRECT=OFF`. Presenting it switches right away; if direct mode is already running with that network, nothing happens – so the card may stay where it is.

More devices on the direct network

- **A phone or notebook** can join the direct network as well. The c64u's web interface is then at `http://192.168.4.64`.
- **A second M5 remote** joins like any other WiFi, most easily with a WiFi card `WIFI:S:C64uRemote-Direct;T:WPA;P:c64ultimate;;`. When it recognises the direct network, it addresses the c64u at `.64` automatically – its home address stays stored. Only **one** device switches direct mode on.

Good to know

- The direct network draws a little more power than normal WiFi operation, because the radio has to stay on all the time.
- The range is fine for a table, less so across a hall.
- *Scan networks* also works in direct mode. During the scan the c64u may briefly lose contact; it reconnects by itself afterwards.

Battery operation and switching off

The M5Dial's battery connector (1.25 mm, 2-pin) takes a single 3.7 V Li-ion/LiPo cell. It is charged whenever the USB cable is plugged in.

Switching off – three ways:

- in the ring menu on *c64u Power Off*, **hold** the button or the icon for **1.5 seconds**,
- *Settings → M5Dial Power Off*, then press a second time within three seconds (*M5DIAL OFF? AGAIN!*),
- present a command card `CMD:M5OFF`.

Once you release the button the M5Dial is off. **Switching on:** press the button.

If the USB cable is plugged in, the device stays powered. It then briefly shows *USB: sleep*, turns the display off and sleeps until you press the button. Pull the cable while it sleeps and the M5Dial is off completely.

The battery does not need a switch of its own. Switched off, the M5Dial only draws a few microamps; a full battery lasts many months that way. If you put the device away for longer than half a year, recharge it once in between.

There is no battery voltage display: the M5Dial has no measuring line for it.

Settings at a glance

Every setting is saved immediately and survives a restart.

NFC tools

Entry	Meaning
NFC-Write	Pick a file and write it to a tag
NFC-Info	Read a tag and show everything about it
NFC-Dump	Save the tag contents to the SD card
NFC-Restore	Write saved contents back to a tag
NFC-Random	Create a random-pick card for a directory
NFC-Cmd	Create a command card (see the *Command cards* chapter)
NFC-Cmd PowOff	Default prompt time for a PowerOff command card: 3, 5, 8, 15 s
WiFi	Wi-Fi setup submenu (see the *Setting up Wi-Fi* chapter)
M5Dial Power Off	Switches the M5Dial itself off - press twice (see the *Battery operation and switching off* chapter)

Operation

Entry	Meaning
Taste lang	Action for a long button press on the home screen
Touch lang	Action for a long touch on the home screen and in the ring menu
PowerOff Abfrage	*Abfrage* = the PowerOff shortcut asks first, *Direkt* = powers off immediately
PowerOff Zeit	Confirmation time window (0.5 to 3.0 s)
Auto-NFC	Background polling interval (*Off*, 1.5s, 0.7s, 0.3s)
Home Timeout	Idle time before returning to the home screen (*Off*, 10 s, 20 s, 45 s)
Encoder Steps	How many detents make one menu step (1, 2, 4, 6)

Taste lang and *Touch lang* can be set to *Off*, *Reset*, *Reboot*, *Menu*, *PowerOff* or *Joy Swap*.

If the menu feels too twitchy when turning, raise *Encoder Steps*.

Display

Entry	Meaning
Animations	Turn the home screen effects on or off (default *Off*)
Effect	*Auto* cycles through all of them, otherwise a fixed one
Anim Speed	Speed of the effects
Effect Time	How long an effect runs
Static Time	How long the still logo shows in between
Brightness	Brightness in eight steps

C64 options

Entry	Meaning
Disk Action	*Mount* only, *Mnt+Reset* adds a reset, *Mnt+Run* also loads and runs
Disk Drive	*Auto (8)* looks for the drive on bus 8, otherwise fixed *A* or *B*

Entry	Meaning
Joystick	Port mapping in the C64: *Normal*, *Swapped*, and depending on firmware *WASD P1* / *WASD P2*

Miscellaneous

Entry	Meaning
Beep	Confirmation tones on or off
Factory Reset	All settings back to defaults (Wi-Fi credentials are kept)

Frequently asked questions

Now and then *Not reached* or *Not verified* shows up and clears again. The HTTP server in the c64u occasionally refuses a connection (“connection refused”) although network and address are fine – this happens even with only a single device on the network. Since v1.2.1 the firmware retries a refused call by itself after a short pause, so you will usually not notice. If it stays that way, restarting the c64u helps.

The c64u keeps dropping out although the M5Dial has good reception. Since v1.4.0 this should no longer happen: the connection to the c64u was reworked thoroughly (see CHANGELOG), and the c64u runs just as reliably on WiFi as on a LAN cable. The main cause was the NFC reader’s permanently switched-on RF field, which disturbed the M5Dial’s WiFi reception. For meetings without a router there is direct mode; there the c64u talks to the M5Dial directly. If the c64u cannot be reached at all after many network changes although its menu shows a connection: unplug its power briefly. Switching it off and on with the button was not enough in that case.

Does the display stutter while the c64u is switched off? Not since v1.3.1. Before, the home screen froze every few seconds for a while as long as the c64u was not on the network. Now the M5Dial checks in the background and shows *Not reached* until the c64u is back. Tip: enter the c64u by its IP address, not by a name - only then does the search run without any interruption.

Touch sometimes does not respond. The whole outer ring is active, so you do not need to hit the icons precisely. If nothing happens at all, check on the *Status* page whether the connection is up – during network trouble the device waits briefly for an answer.

The tag is not detected on its own. Background polling only runs on the home screen and in the ring menu, not in the settings or the file browser. Also check whether *Settings → Auto-NFC* is set to *Off*.

Turning on the home screen only opens the menu. That is intentional: the first turn opens the menu only, so you do not accidentally skip past a menu entry.

The SD card is not detected. Format the card as FAT32, check the wiring and especially the supply voltage of the module (see the technical documentation), and keep the Grove cables short.

Uploading a .d64 takes a while. A disk image is about 175 kB and travels over Wi-Fi; a few seconds are normal. Loading inside the C64 can take a little longer afterwards – the progress is shown on the display.

A tag is read but nothing starts. Usually the path on the tag does not match the SD card. *NFC-Info* tells you whether the file was found.

Can I use a tag from my TeensyROM? Yes. The tag format is identical. The folder and file names on your SD card just have to match.

Are my settings kept? Yes, they live in the flash memory of the M5Dial. Only *Factory Reset* clears them.

License

C64uRemote is released under the **MIT License**. The full text is in the file `LICENSE` in the project root.

It originates from **C64uRemote by Karl Prosser (@klumsy)**, https://github.com/ReadyOS-C64/C64uRemote, which he published under the MIT License. This version is a derivative work and is released under the same terms:

- Copyright (c) 2026 Karl Prosser – original project
- Copyright (c) 2026 Martin Oswald (@mad, https://1MHz.de) – port and extensions

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The libraries used carry their own licenses: M5Unified and M5GFX (MIT, © M5Stack), ArduinoJson (MIT, © Benoit Blanchon) and MFRC522_I2C (https://github.com/kkloesener/MFRC522_I2C). PlatformIO fetches them at build time.

How this was made

The port, the extensions and these manuals were written with the help of Claude (Anthropic). Concept, idea, hardware decisions and every test on real devices: Martin Oswald (@mad).